

Question 1 (11 Marks)**Marks**

- a) Solve for x : $\frac{x-2}{x+3} + 2 \geq 0$ 3
- b) Simplify $\frac{y}{x^2 - xy} + \frac{1}{x}$ 2
- c) Find the acute angle between the lines $5y = 3x + 1$ and $4x - y = 3$ 2
- d) Suppose that P is the point $(-4, 7)$ and Q is the point $(1, -3)$.
- (i) Find the point R which divides the interval PQ internally in the ratio $c:1$. 1
- (ii) Hence, or otherwise, find the ratio in which the line $3x + 4y = 6$ divides the interval PQ. 3

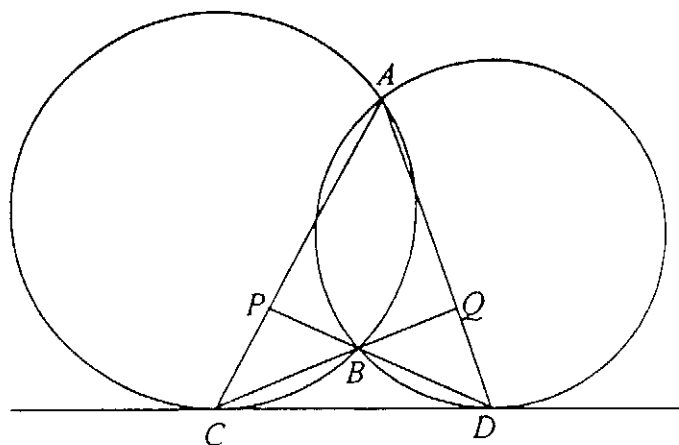
Question 2 (11 Marks) Start a new page

- a) What are the co-ordinates of the focus of the parabola $x^2 + 6x + 8y - 7 = 0$ 2
- b) If α and β are the roots of the equation $4x^2 - 2x - 1 = 0$ find (without solving for x)
- (i) $\frac{1}{2\alpha} + \frac{1}{2\beta}$ 3
- (ii) $\alpha^2 + \beta^2$ 1
- (iii) $\alpha - \beta$ 2
- c) What value(s) of k will make the expression $(k+1)x^2 - 2(k-1)x + (2k-5)$ a perfect square? 3

Question 3 (10 Marks) Start a new page

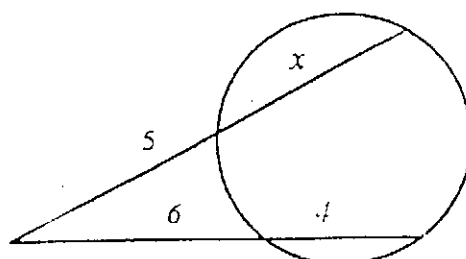
Marks

a)



Two circles intersect at A and B . A common tangent touches both circles at C and D , as shown in the diagram above. The line DB meets the chord AC at P , and the line CB meets the chord AD at Q .

- (i) Make a large, neat copy of the diagram on your answer sheet, and draw the common chord AB . 1
 - (ii) Let $\angle BCD = \alpha$ and $\angle BDC = \beta$. 1
Give a reason why $\angle CAB = \alpha$ and $\angle DAB = \beta$.
 - (iii) Show that $\angle PBQ = 180^\circ - (\alpha + \beta)$. 2
 - (iv) Give a reason why $APBQ$ is a cyclic quadrilateral. 1
 - (v) Show that $\angle PQB = \alpha$. 2
 - (vi) Hence show that PQ is parallel to CD . 1
- b) Find the value of x in the diagram below. 2



Question 4 (10 Marks) Start a new page

Marks

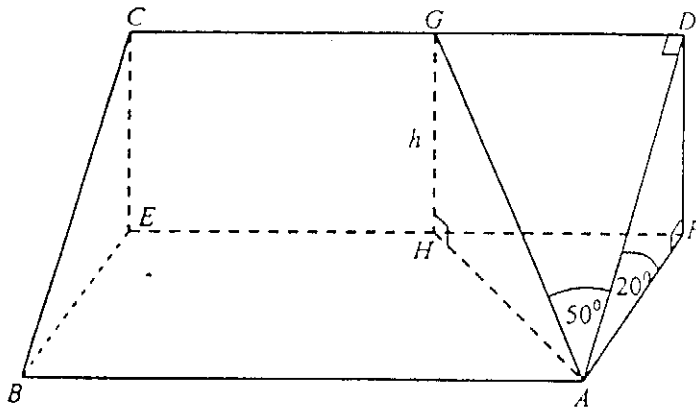
a) Show that $\frac{\cos \theta}{1 - \sin \theta} - \sec \theta = \tan \theta$.

3

b) Solve for $0^\circ \leq x \leq 360^\circ$: $\sin^2 2x = \frac{1}{4}$

3

- c) A plane hillside $ABCD$ makes an angle of 20° with the horizontal.
A path AG makes an angle of 50° with a line of greatest slope.
If $DF = GH = h$:



(i) Show that $AD = \frac{h}{\sin 20^\circ}$

1

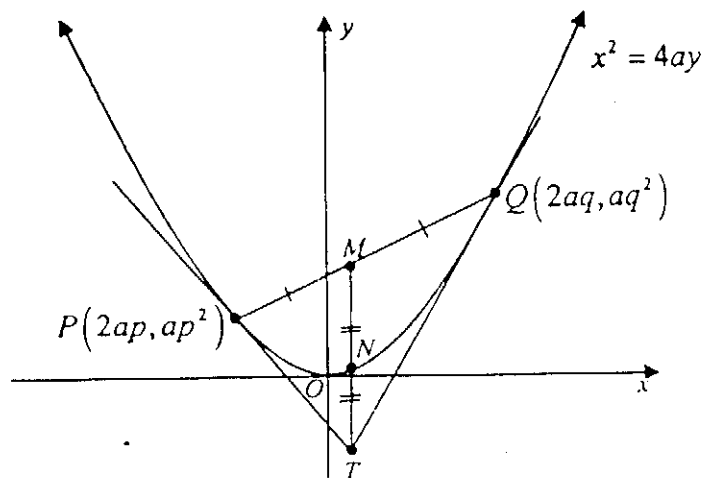
(ii) Show that $AG = \frac{h}{\sin 20^\circ \cos 50^\circ}$

1

- (iii) Hence, find the inclination of the path AG to the horizontal.
(Leave your answer to the nearest degree)

2

a)



In the diagram, $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ are two points on the parabola $x^2 = 4ay$.

(i) Show that the tangent at P has equation $y = px - ap^2$. 2

(ii) The tangents at P and Q meet at T . Assuming that the tangent at Q is $y = qx - aq^2$, show that T is the point $(a(p+q), apq)$. 2

(iii) M is the midpoint of the chord PQ . Show that MT is parallel to the axis of symmetry of the parabola. 2

b) Find $\lim_{x \rightarrow -5} \frac{\sqrt{20-x} - 5}{5+x}$ 3

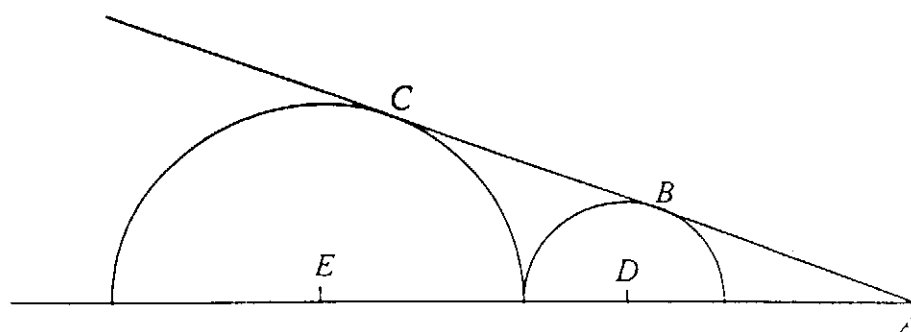
c) If the roots of the equation $4(p^2 + q^2)x^2 + 4prx + (r^2 - 4q^2) = 0$ are real and $q \neq 0$, then show that 3

$$p^2 + q^2 \geq \frac{r^2}{4}$$

Question 6 (10 Marks) Start a new page

Marks

a)



The diagram above shows two semicircles centred at D and E with radii 1 cm and 3 cm respectively. ABC is a common tangent to both semicircles.

(i) Find the length of AD 4

(ii) Find the size of $\angle DAB$ 1

(iii) Find the length of AB 1

b) (i) Show that $\frac{12}{4-x} - 2 = \frac{2x+4}{4-x}$ 1

(ii) Hence, or otherwise sketch $y = \frac{2x+4}{4-x}$ 2

(iii) Hence, or otherwise solve for x : $\frac{2x+4}{4-x} \geq 0$ 1

End of Paper

2002 - AP2 EXTENSION

SOLUTIONS

1. a) $\frac{x-2}{x+3} \geq -2$

$$(x-2)(x+3) \geq -2(x+3)^2 \quad (1)$$

$$(x-2)(x+3) + 2(x+3)^2 \geq 0$$

$$(x+3)(x-2+2x+6) \geq 0$$

$$(x+3)(3x+4) \geq 0 \quad (1)$$

$$\therefore x < -3 \text{ or } x \geq -\frac{4}{3} \quad (1)$$

b) $\frac{xy + x^2 - xy}{x(x^2 - xy)} \quad (1)$

$$= \frac{x^2}{x(x^2 - xy)}$$

$$= \frac{x^2}{x^2(x-y)}$$

$$= \frac{1}{x-y} \quad (1)$$

c) $3x - 5y + 1 = 0 \quad (1)$

$$4x - y - 3 = 0 \quad (2)$$

$$m_1 = \frac{3}{5} \quad m_2 = 4 \quad (1)$$

$$\therefore \tan \theta = \left| \frac{\frac{3}{5} - 4}{1 + \frac{3}{5} \times 4} \right|$$

$$= \frac{\frac{17}{5} \times \frac{5}{17}}{1} = 1$$

$$\theta = 45^\circ \quad (1)$$

d) i) R: $\left(\frac{c-4}{1+c}, \frac{-3c+7}{1+c} \right)$

$$= \left(\frac{c-4}{c+1}, \frac{7-3c}{c+1} \right) \quad (1)$$

ii) $3\left(\frac{c-4}{c+1}\right) + 4\left(\frac{7-3c}{c+1}\right) = 6 \quad (1)$

$$3c - 12 + 28 - 12c = 6c + 6 \quad (1)$$

$$10 = 9c$$

$$c = \frac{10}{9} = \frac{2}{3} \quad (1)$$

$$\therefore \text{Ratio is } \frac{10}{9} : \frac{2}{3} = 5 : 3$$

2. a) $x^2 + 6x + 9 = 7 - 8y + 9$

$$(x+3)^2 = 16 - 8y \quad (1)$$

$$= -8(y-2)$$

$$\therefore \text{Focus} = (-3, 0) \quad (1)$$

b) i) $\alpha + \beta = \frac{1}{2}, \alpha\beta = -\frac{1}{4} \quad (1)$

$$\frac{1}{2\alpha} + \frac{1}{2\beta} = \frac{2(\alpha+\beta)}{4\alpha\beta}$$

$$= \frac{\alpha+\beta}{2\alpha\beta} \quad (1)$$

$$= \frac{1}{2} \times \frac{-2}{1}$$

$$= -1 \quad (1)$$

ii) $\alpha^2 + \beta^2 = (\alpha+\beta)^2 - 2\alpha\beta$

$$= \frac{1}{4} + \frac{1}{2}$$

$$= \frac{3}{4} \quad (1)$$

$$(ii) \alpha - \beta = \pm \sqrt{(\alpha + \beta)^2 - 4\alpha\beta} \quad (1)$$

$$= \pm \sqrt{\frac{1}{4} + 1}$$

$$= \pm \frac{\sqrt{5}}{2} \quad (1)$$

(vii) $\therefore PQ \parallel CD$ (alternate Δ 's are equal) (1)

b) $5(x+5) = 6 \times 10$ (product of intercepts are equal) (1)

$$5x + 25 = 60$$

$$x = 7 \quad (1)$$

c) $\Delta = 0$

$$4(k-1)^2 - 4(k+1)(2k-5) = 0 \quad (1)$$

$$(k-1)^2 - (k+1)(2k-5) = 0$$

$$k^2 - 2k + 1 - 2k^2 + 3k + 5 = 0$$

$$-k^2 + k + 6 = 0$$

$$k^2 - k - 6 = 0$$

$$(k-3)(k+2) = 0$$

$$\therefore k = +3 \text{ or } -2 \quad (1)$$

But $k \neq -2$ as $a < 0$

$$\therefore k = 3 \quad (1)$$

4. a) $\angle HS = \frac{\cos \theta}{1 - \sin \theta} - \frac{1}{\cos \theta}$

$$= \frac{\cos^2 \theta - 1 + \sin \theta}{\cos \theta (1 - \sin \theta)} \quad (1)$$

$$= \frac{1 - \sin^2 \theta - 1 + \sin \theta}{\cos \theta (1 - \sin \theta)} \quad (1)$$

$$= \frac{\sin \theta - \sin^2 \theta}{\cos \theta (1 - \sin \theta)}$$

$$= \frac{\sin \theta (1 - \sin \theta)}{\cos \theta (1 - \sin \theta)} \quad (1)$$

$$= \tan \theta$$

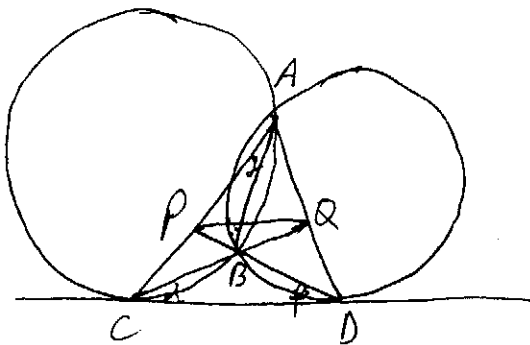
$$= RHS$$

b) $\sin 2x = \pm \frac{1}{2} \quad (1)$

$$\therefore 2x = 30^\circ, 150^\circ, 210^\circ, 330^\circ, 390^\circ, 510^\circ, 570^\circ, 690^\circ \quad (1)$$

$$x = 15^\circ, 75^\circ, 105^\circ, 165^\circ, 195^\circ, 255^\circ, 285^\circ, 345^\circ \quad (1)$$

30/12



(1)

(i) Alternate segment theorem (1)

(ii) $\angle CBD = 180 - (\alpha + \beta)$ (Δ sum of Δ) (1)

$\therefore \angle BQ = 180 - (\alpha + \beta)$ (vertically opposite Δ) (1)

(iii) Opposite Δ 's are supplementary (1)

(iv) Since APBQ is a cyclic quad (1)

$\angle PAB = \angle$ (Δ 's in same segment) (1)

c) (i) $DP = h$

$$\therefore \frac{h}{AD} = \sin 20$$

$$h = AD \sin 20 \quad (1)$$

$$AD = \frac{h}{\sin 20}$$

$$(ii) \frac{AD}{AG} = \cos 50^\circ$$

$$AG = \frac{AD}{\cos 50^\circ} \quad (1)$$

$$= \frac{h}{\sin 20 \cos 50}$$

$$(iii) \frac{h}{AG} = \sin \theta$$

$$\therefore \frac{AG \sin 20 \cos 50}{AG} = \sin \theta \quad (1)$$

$$\therefore \sin \theta = \sin 20 \cos 50$$

$$\theta \approx 13^\circ \quad (1)$$

$$5. (i) y = \frac{x^2}{4a}$$

$$\frac{dy}{dx} = \frac{2x}{4a}$$

$$= \frac{x}{2a} \quad (1)$$

$$\text{At } P \frac{dy}{dx} = p$$

$$\therefore y - ap^2 = p(x - 2ap) \quad (1)$$

$$= px - 2ap^2$$

$$y = px - ap^2$$

$$(ii) y = px - ap^2$$

$$y = qx - aq^2$$

$$(p - q)x = a(p^2 - q^2) \quad (1)$$

$$x = a(p + q)$$

$$y = ap(p + q) - ap^2$$

$$= ap^2 + apq - ap^2$$

$$= apq \quad (1)$$

$$\therefore T \equiv (a(p + q), apq)$$

$$M = \left(\frac{2ap + 2aq}{2}, \frac{ap^2 - aq^2}{2} \right)$$

$$= \left(a(p + q), \frac{a(p^2 - q^2)}{2} \right) \quad (1)$$

Since M has same x co-ordinate as T, MT is vertical. (1)

$\therefore$ MT is parallel to axis of parabola

$$b) = \lim_{x \rightarrow -5} \frac{\sqrt{20-x} - 5}{5+x} \cdot \frac{\sqrt{20-x} + 5}{\sqrt{20-x} + 5} \quad (1)$$

$$= \lim_{x \rightarrow -5} \frac{20-x-25}{(5+x)(\sqrt{20-x} + 5)}$$

$$= \lim_{x \rightarrow -5} \frac{-5-x}{(5+x)(\sqrt{20-x} + 5)}$$

$$= \lim_{x \rightarrow -5} \frac{-1}{\sqrt{20-x} + 5} \quad (1)$$

$$= -\frac{1}{10} \quad (1)$$

$$c) \Delta \geq 0$$

$$\therefore 16p^2r^2 - 16(p^2 + q^2)(r^2 - 4q^2) \geq 0 \quad (1)$$

$$16p^2r^2 - 16(p^2r^2 - 4p^2q^2 + q^2r^2 - 4q^4) \geq 0$$

$$64p^2q^2 - 16q^2r^2 + 64q^4 \geq 0 \quad (1)$$

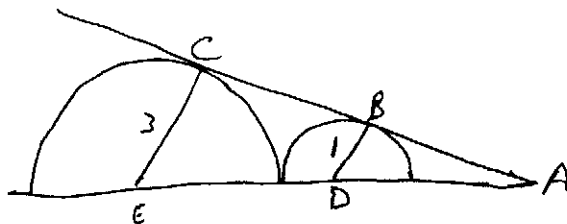
$$4p^2q^2 - q^2r^2 + 4q^4 \geq 0$$

$$4p^2q^2 + 4q^4 \geq q^2r^2$$

$$4q^2(p^2 + q^2) \geq q^2r^2 \quad (1)$$

$$p^2 + q^2 \geq \frac{r^2}{4}$$

6. a) (i)



∴ $\Delta ABD \sim \Delta ACE$

$\hat{A}$ is common

$\hat{DBA} = \hat{ECB} = 90^\circ$ (4 between tangent & radius) ①

∴ $\Delta ABD \parallel \Delta ACE$ (equiangular) ①

$$\frac{AD}{AE} = \frac{1}{3} \quad (\text{sides in proportion})$$

$$\frac{AD}{AD+4} = \frac{1}{3} \quad \text{①} \rightarrow$$

$$3AD = AD+4$$

$$2AD = 4$$

$$AD = 2 \quad \text{①}$$

$$(ii) \sin \hat{DAB} = \frac{1}{2}$$

$$\therefore \hat{DAB} = 30^\circ \quad \text{①}$$

$$(iii) AB^2 = 2^2 - 1^2$$

$$AB = \sqrt{3} \text{ cm} \quad \text{①}$$

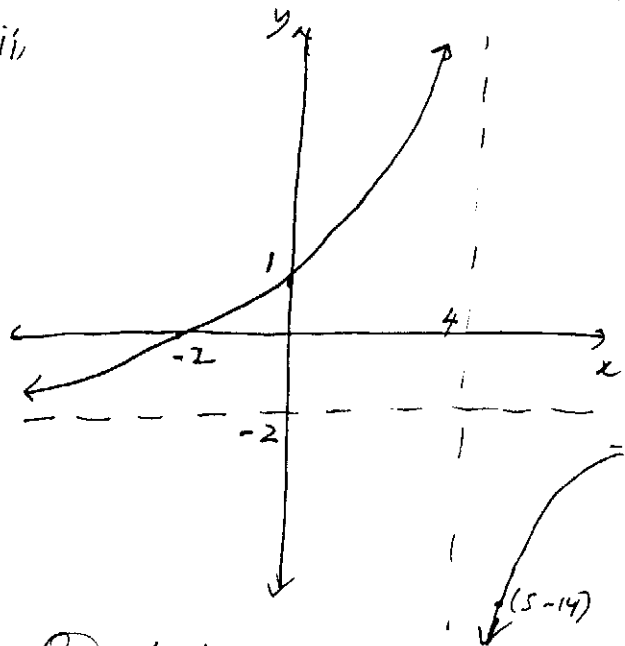
$$b) (i) LHS = \frac{12 - 2(4-x)}{4-x}$$

$$= \frac{12 - 8 + 2x}{4-x}$$

$$= \frac{4 + 2x}{4-x} \quad \text{①}$$

$$= RHS$$

(ii)



② 1 shape
1 asymptote and axes

$$(iii) -2 \leq x < 4 \quad \text{①}$$